



ESCUELA DE INGENIERÍA
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Lecture 23: Oligopoly II

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ICS2523 Microeconomics

Lecture 23

Last lecture, we started studying Oligopoly:

- Definitions: Best response, Nash Equilibrium, dominant strategy
- Review examples of matrix games

In today's lecture, we will continue studying **Oligopoly**:

- Review more examples of matrix games
- Backward induction
- Example of continuum of actions → Tragedy of the Commons
- Oligopoly models: Bertrand, Cournot, Cartel, Stackelberg

Game Theory

definitions

Market Failures

Example Rock, Paper, Scissors

Rock, Paper, Scissors (*Cachipún*) is a children's game in which the two players simultaneously display one of three hand symbols

	Rock	Paper	Scissors
Rock	(0,0)	(-1, <u>1</u>)	(<u>1</u> ,-1)
Paper	(<u>1</u> ,-1)	(0,0)	(-1, <u>1</u>)
Scissors	(-1, <u>1</u>)	(<u>1</u> ,-1)	(0,0)

In this example (x,y) represent positive payoffs

None of the nine boxes represents a Nash Equilibrium (in pure strategies, there is one N.E. in mixed strategies)

Any strategy pair is unstable because it offers at least one of the players an incentive to deviate.

8. Game Theory definitions

You may see a real game theory game on TV:

<https://www.youtube.com/watch?v=S0qjK3TWZE8>

- **Sequential games:** In some games, the order of moves matters. The big difference with simultaneous games is that a player that moves later in the game can observe how others have played up to that moment
 - Player can use this information to form more sophisticated strategies than simply choosing an action → he may have a contingent plan (with the action played depending on what the others have done)
- **Backward induction:** process of solving the equilibrium by working backwards (from the end of the game to the beginning)

8. Game Theory definitions

- **Repeated games:** In many real-world settings, players play the same game over and over again.
 - The simple constituent game that it is played repeatedly is called the *stage game*.
 - Repeated play of the stage game opens up the possibility of cooperation in equilibrium
 - Players can adopt ***trigger strategies*** (they cooperate as long as all have cooperated, but revert to playing the Nash equilibrium if anyone deviates from cooperation)
- **Finitely versus Infinitely** repeated games
 - Finitely repeated games: if the stage game has a unique Nash equilibrium, then the unique subgame-perfect equilibrium of the finitely repeated game is to play the Nash equilibrium every period (Selten's Theorem). If the stage game has multiple Nash equilibria, it may be possible to achieve some cooperation
 - Infinitely repeated games: Players may end up cooperating

8. Game Theory definitions

- **Incomplete information:** some players have information about the game that others do not
 - Examples?
 - These are complicated games, players that lack full information about the game will try to use what they do know to make **inferences** about what they do not → Bayesian games (use of bayesian rule)
 - Incomplete information games can also be sequential → player 1 moves first (*signaling games*)
- **Continuum of actions:** some settings are more realistically modeled through a continuous range of actions. We will see this further with Bertrand (players select a positive price) and Cournot (players select a positive quantity) models.



Continuum of actions: Example

The Tragedy of the Commons

Two herders decide how many sheep to graze on the village commons

Problem: commons is quite small and can rapidly succumb to overgrazing

Continuum of actions: Example

The Tragedy of the Commons

Let q_i be the number of sheep that herder $i = 1, 2$ grazes on the commons

Per-sheep value of grazing on the commons (in terms of wool and sheep-milk) is: $v(q_1, q_2) = 120 - (q_1 + q_2)$

First step: write down the payoff for each player as a function of all player's actions

$$u_1(q_1, q_2) = q_1 v(q_1, q_2) = q_1(120 - (q_1 + q_2))$$

$$u_2(q_1, q_2) = q_2 v(q_1, q_2) = q_2(120 - (q_1 + q_2))$$

Continuum of actions: Example

The Tragedy of the Commons

Second step: compute the first-order condition associated with each player's payoff maximum $\rightarrow$ find the **best response of each player** as a function of all other player's actions

N players $\rightarrow$ n equations for the n unknown equilibrium actions.

To find the Nash equilibrium we solve Herder 1's value-maximization problem

$$\max_{\{q_1\}} q_1(120 - (q_1 + q_2))$$

Continuum of actions: Example

The Tragedy of the Commons

$$\max_{\{q_1\}} q_1(120 - (q_1 + q_2))$$

First order condition:

$$120 - 2q_1 - q_2 = 0 \rightarrow q_1 = 60 - q_2/2 = BR_1(q_2) \text{ (herder 1's best response)}$$

Similarly for herder 2: $\rightarrow q_2 = 60 - q_1/2 = BR_2(q_1)$ (herder 2's best response)

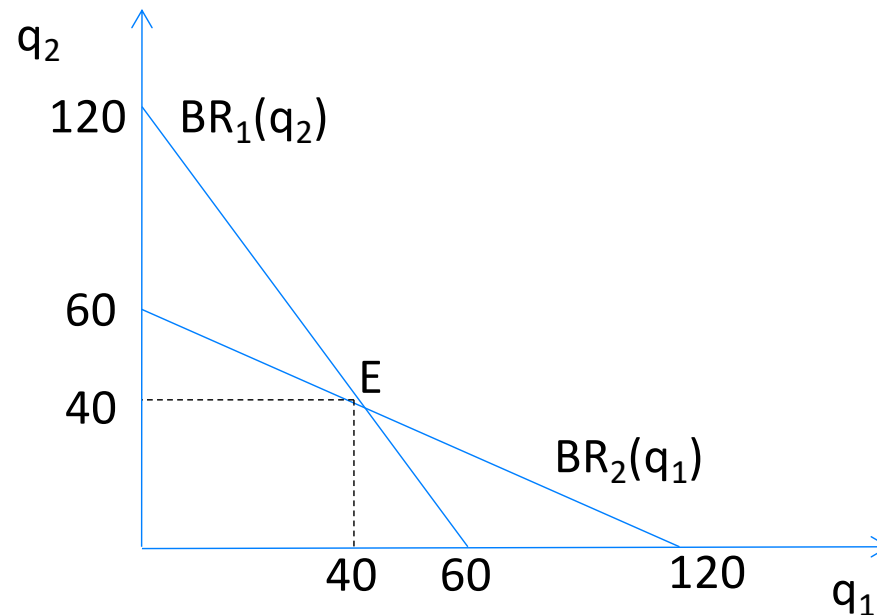
The Nash equilibrium is given by the pair (q_1^*, q_2^*) that solve the best response equations $\rightarrow (40, 40)$ and each herder earns a payoff of 1,600

PROBLEM: Nash eq is NOT the best use of the commons (less efficient)

Continuum of actions: Example

The Tragedy of the Commons

PROBLEM: Nash eq is NOT the best use of the commons (less efficient)



Joint Payoff maximization $\rightarrow (30, 30)$ gives 1,800 each

Continuum of actions: Example

The Tragedy of the Commons

Joint payoff maximization

$$\max_{\{q_1\}} (q_1 + q_2)(120 - (q_1 + q_2))$$

First order condition:

$$(120 - q_1 - q_2) - q_1 - q_2 = 0 \rightarrow q_1 = 60 - q_2 = BR_1(q_2) \text{ (herder 1's best response)}$$

Similarly for herder 2: $\rightarrow q_2 = 60 - q_1 = BR_2(q_1)$ (herder 2's best response)

This joint Payoff maximization is solved by $q_1 = q_2 = 30$ or, more generally, by any q_1 and q_2 that sum to 60. Each herder earns a payoff of 1,800. Nevertheless, this does not happen naturally.

Backward Induction

Market Failures

13. Backward Induction

Backward induction is the process of solving for equilibrium by working backward from the end of the game to the beginning.

Backward induction works as follows:

1. **Identify all the subgames** at the bottom of the extensive form
2. **Find the Nash equilibria** on these subgames
3. **Replace** the (potentially complicated) subgames with the **actions and payoffs resulting from the Nash equilibrium** play on these subgames
4. Then, **move up to the next level of subgames** and repeat the procedure

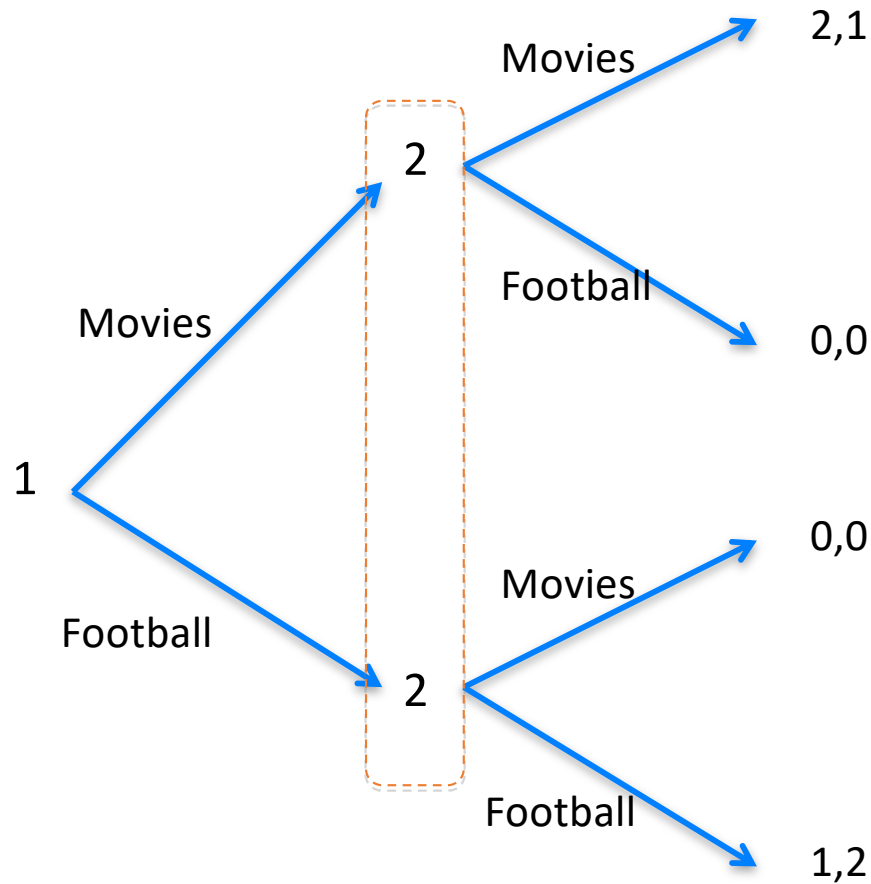
What is a subgame?

A **subgame** is a part of the extensive form beginning with a decision node and including everything that branches out to the right of it.

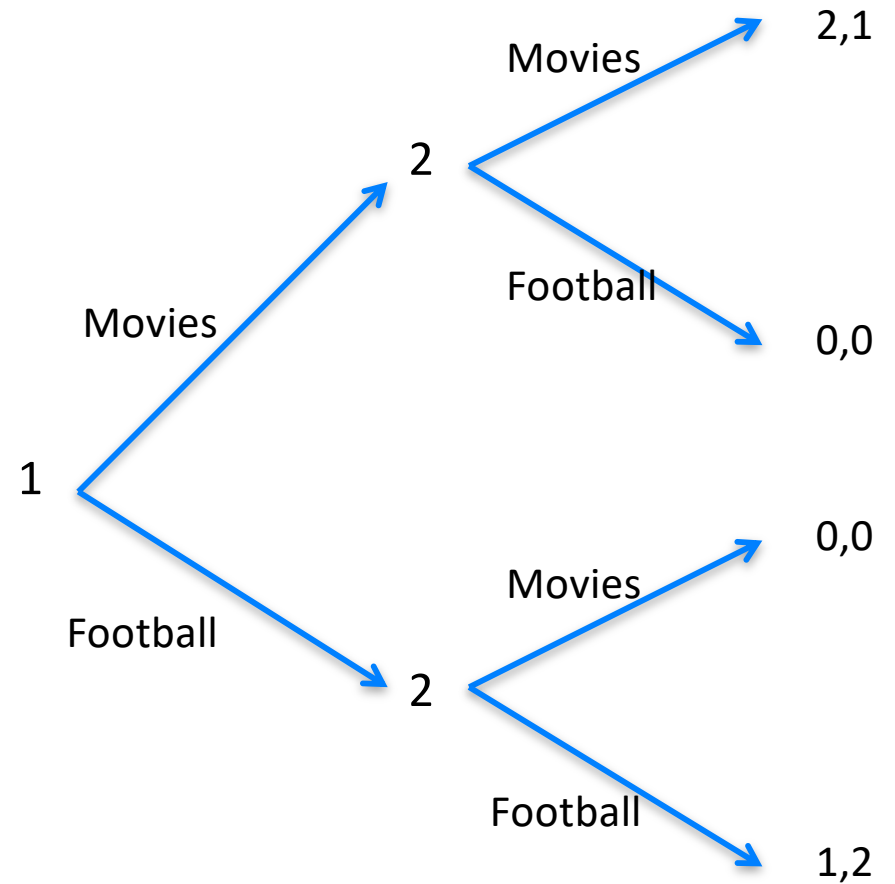
This means that a player that moves first knows the actions played by the others that have led to that point.

Let's review the Battle of the Sexes from a sequential and simultaneous version.

Battle of the Sexes extensive form



SIMULTANEOUS



SEQUENTIAL

Player 1: wife, Player 2: husband

Activity 1

Identify the subgames present in the Battle of the Sexes example depicted in the previous slide, for its simultaneous and sequential versions.

13. Backward Induction

Backward induction is particularly useful in games that feature many rounds of sequential play. As rounds are added, it quickly becomes too hard to solve for all the Nash equilibria and then to sort through which are subgame – perfect.

With backward induction, an additional round is simply accommodated by adding another iteration of the procedure.

Oligopoly models

Market Failures

Oligopoly models

Now that we know some definitions of Game Theory, we can move on to modelling Oligopoly.

We will study four models of Oligopoly

1. **Bertrand** model (price)
2. **Cournot** model (quantity)
3. **Cartel** model (cooperative)
4. **Stackelberg** model (sequential)

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Bertrand Model

Market Failures

9. Oligopoly: Bertrand Model

It is a duopoly with two identical firms **choosing prices** simultaneously for their identical products. Outputs are produced at a constant marginal cost and average cost (c).

Firms choose prices P_1 and P_2 , simultaneously, in a single period of competition.

Firm's **products are perfect substitutes**, so all sales goes to the firm with the lowest price or they split evenly if $P_1 = P_2$

$D(P)$ will be the market demand. The game has a continuum of actions

Activity 2

Verify that $P_1^* = P_2^* = c$ is a Nash equilibrium and there is no other one.

HINT: you have to show that both firms are playing a **best response** to each other (neither of them has an incentive to deviate to some other strategy)

9. Oligopoly: Bertrand Model

If $D(P) = a - P$, for example (where $a > c$), then the demand curve of firm 1 will be:

$$q_1(P_1, P_2) = \begin{cases} 0 & , \text{if } P_1 > P_2 \\ 0,5 \cdot Q(P_1) & , \text{if } P_1 = P_2 \\ Q(P_1) & , \text{if } P_1 < P_2 \end{cases}$$

Thus,

$$\pi_1(P_1, P_2) = \begin{cases} 0 & , \text{if } P_1 > P_2 \\ P_1 \cdot 0,5 \cdot Q(P_1) - c \cdot 0,5 \cdot Q(P_1) & , \text{if } P_1 = P_2 \\ P_1 \cdot Q(P_1) - c \cdot Q(P_1) & , \text{if } P_1 < P_2 \end{cases}$$

9. Oligopoly: Bertrand Model

Warning: Be careful **not to do** something like this:

$$\begin{aligned} \text{Max}_P \pi_1(P) &= P \cdot 0.5 \cdot Q(P) - c \cdot 0.5 \cdot Q(P) \\ &= 0.5 \cdot (P - c) \cdot (a - P) \end{aligned}$$

First order condition: $P^* = \frac{a+c}{2}$

Analogously, for firm 2 we would reach the same “optimality” condition $P^* = \frac{a+c}{2}$

Is this result ($P_1^* = P_2^* = \frac{a+c}{2}$) a Nash equilibrium?

Why couldn't we use the first-order conditions?

9. Oligopoly: Bertrand Model

Since the profit function of each firm is a discontinuous function, we cannot use differential calculus.

Bertrand paradox: Nash equilibrium of the Bertrand model is the same as the perfectly competitive outcome!

Paradoxically, competition between two firms could be so tough. It is a general result in the sense that we did not specify the marginal cost c or the demand curve (it should be downward-sloping)



Activity 3

Two firms produce identical cat food with constant marginal costs of C_1 and C_2 , respectively, with $C_1 < C_2$.

What is the Bertrand equilibrium in this model?

Cournot Model

Market Failures

10. Oligopoly: Cournot Model

Instead of choosing prices, **firms choose quantities**. This small change in strategic variable will lead to a big change in implications

We will have n firms $\rightarrow i = 1, 2, \dots, n$

Each firm chooses its output q_i of an identical product simultaneously

10. Oligopoly: Cournot Model

Total industry output: $Q = q_1 + q_2 + \dots + q_n$ resulting in market price $P(Q)$

(inverse demand curve, demand curve $Q = D(P)$).

We will assume demand curve is downward sloping $\rightarrow P'(Q) < 0$

In this case, profit function is differentiable:

$$\pi_i = P(Q)q_i - C_i(q_i)$$

We can find firm's i best response by looking at the first-order condition:

$$\frac{\partial \pi_i}{\partial q_i} = P(Q) + P'(Q)q_i - C_i'(q_i) = 0$$

10. Oligopoly: Cournot Model

$$\frac{\partial \pi_i}{\partial q_i} = P(Q) + P'(Q)q_i - C_i'(q_i) = 0$$

This condition must hold for all $i=1, 2, \dots, n$ in the Nash equilibrium

MR= MC

Price must exceed MC by the magnitude of the term $P'(Q)q_i$

Price will be above the perfectly competitive level (above MC) but below the level in a perfect cartel that maximizes firms' joints profits

Let's compare this outcome with a perfect cartel outcome:

10. Oligopoly: Cournot Model

Cournot Example

For a duopoly with identical firms (and with a constant marginal cost equal to c), facing a total demand given by $(P(Q)=a-Q)$:

Firm 1 will try to maximize its profits:

$$\begin{aligned} \text{Max } \pi_1(q_1, q_2) &= P(q_1 + q_2) \cdot q_1 - c \cdot q_1 \\ &= (a - q_1 - q_2) \cdot q_1 - c \cdot q_1 \end{aligned}$$

First order condition:

$$a - q_1 - q_2 - q_1 - c = 0$$

$$q_1 = 0.5 \cdot (a - c - q_2) \text{ Firm 1 Best-response}$$

Similarly, for Firm 2:

$$q_2 = 0.5 \cdot (a - c - q_1) \text{ Firm 2 Best-response}$$

10. Oligopoly: Cournot Model

Cournot Example

Thus, the only Nash equilibrium for this game is:

$$q_1^* = \frac{a - c}{3}$$

$$q_2^* = \frac{a - c}{3}$$

The equilibrium price will be: $P^* = P(q_1^*, q_2^*) = \frac{a + 2 \cdot c}{3}$ and the profit for each firm will be:

$$\pi_1^* = P^* \cdot q_1^* - c \cdot q_1^* = \frac{(a - c)^2}{9}$$

$$\pi_2^* = P^* \cdot q_2^* - c \cdot q_2^* = \frac{(a - c)^2}{9}$$

Individual Activity

On a piece of paper, write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it in your own words.

If you have any **questions** about what we discussed today, write them down too. We Will go through them at the beginning of the next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.